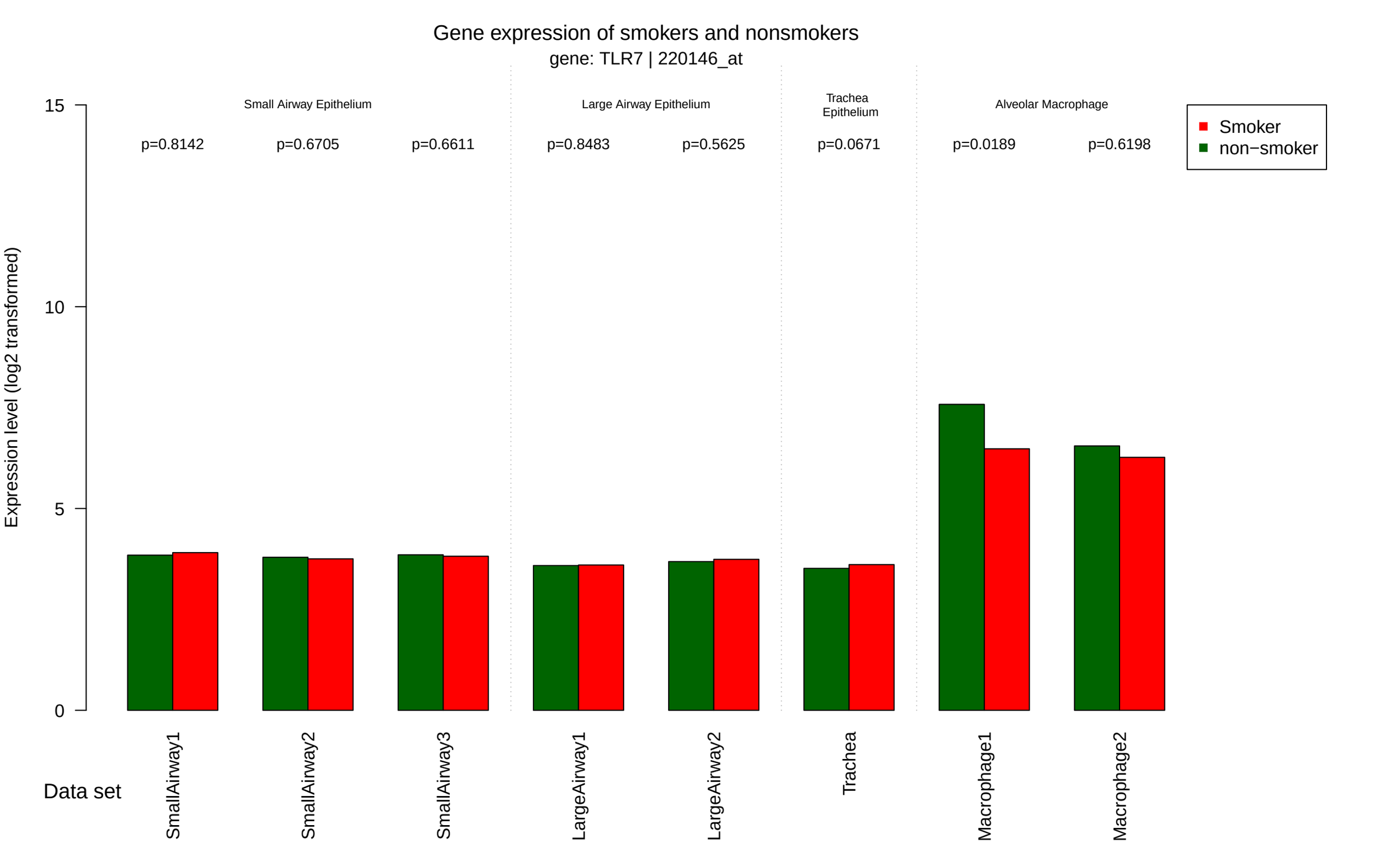
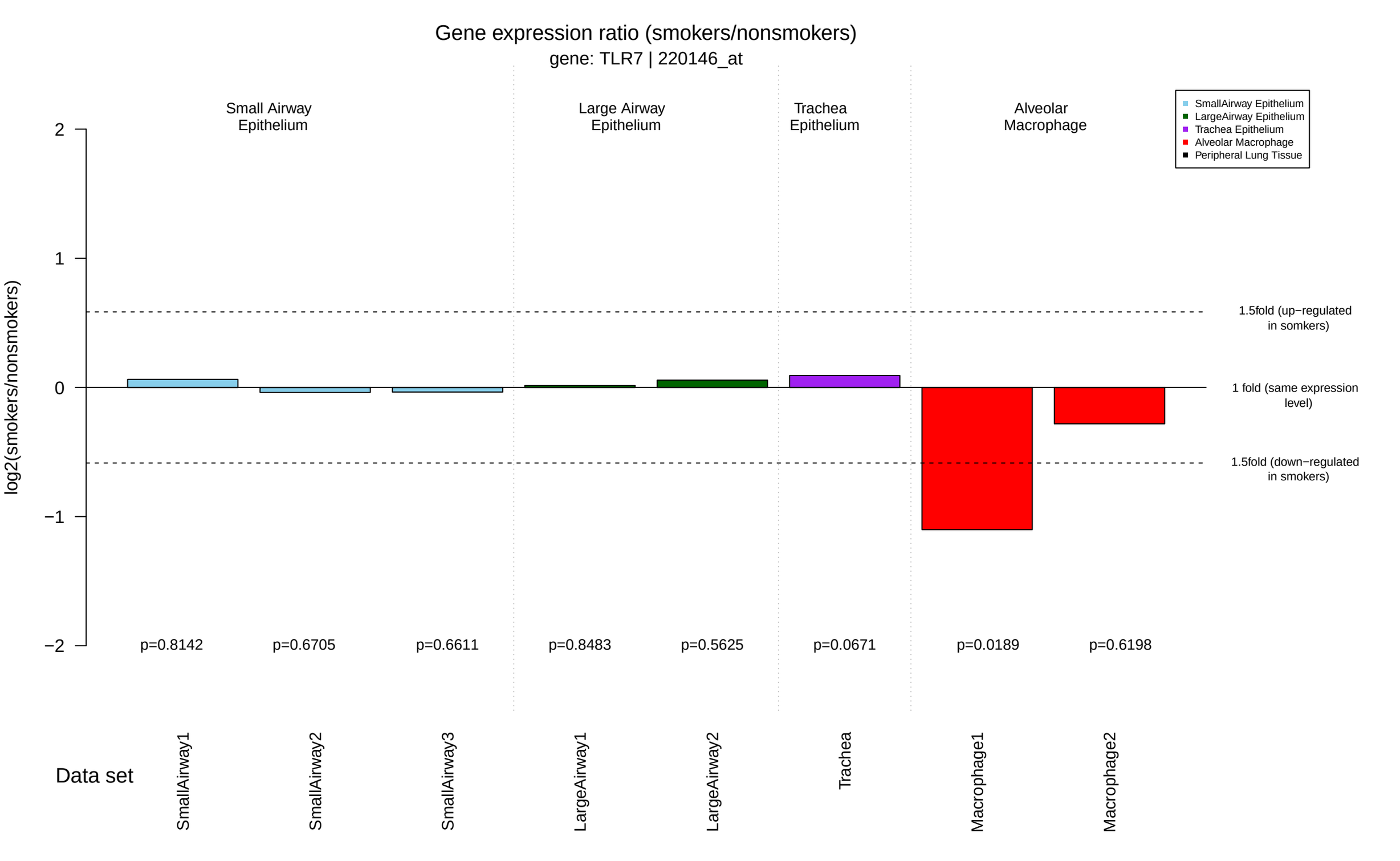
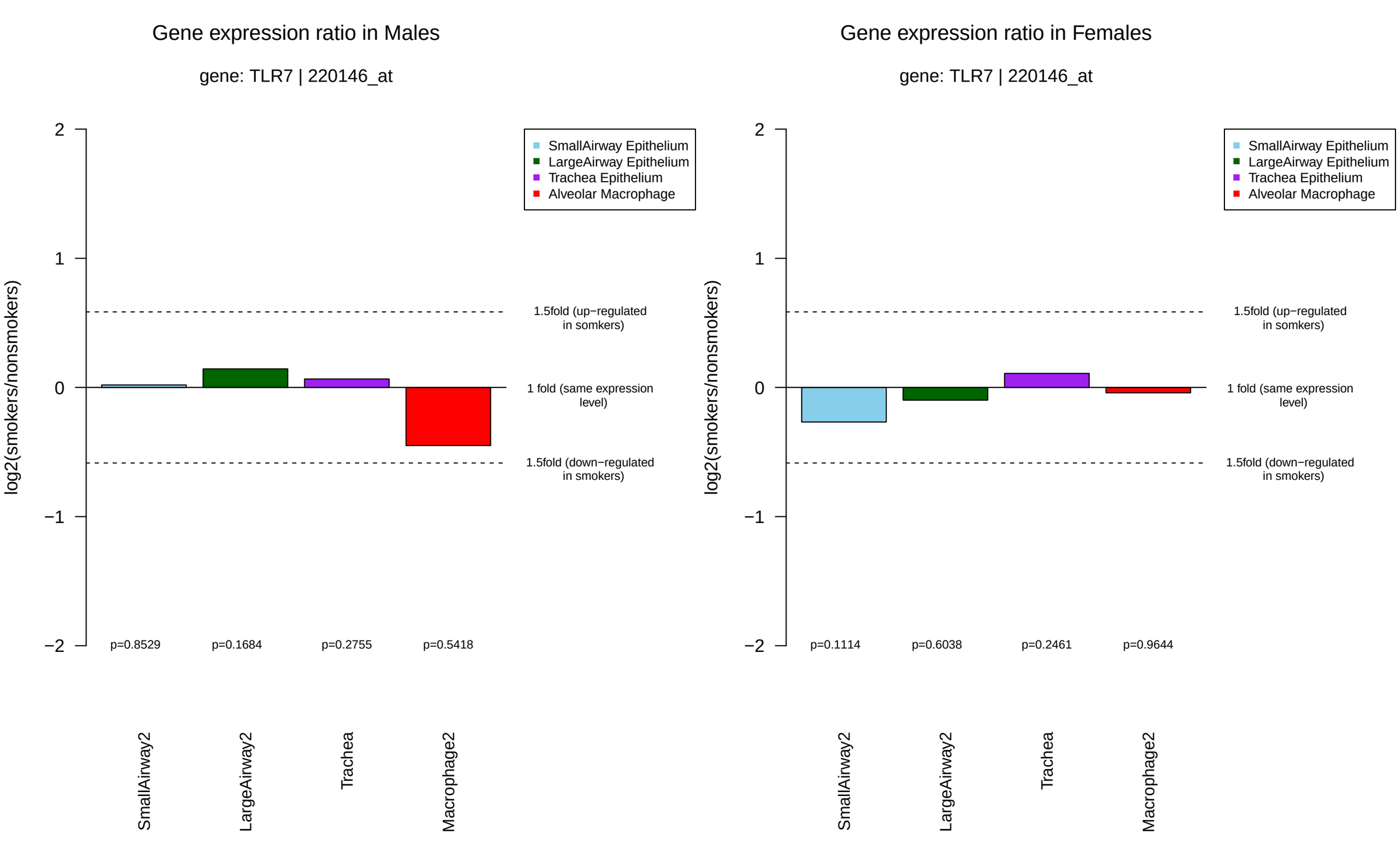
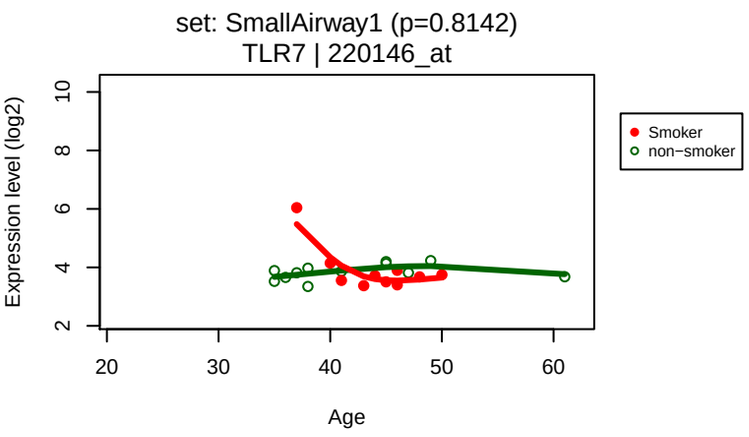
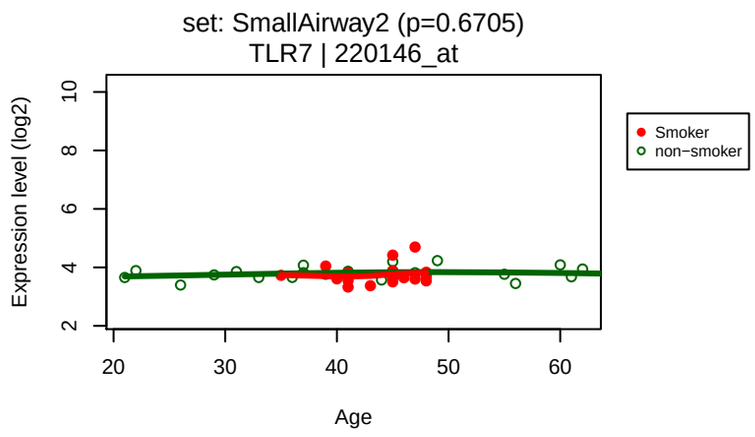
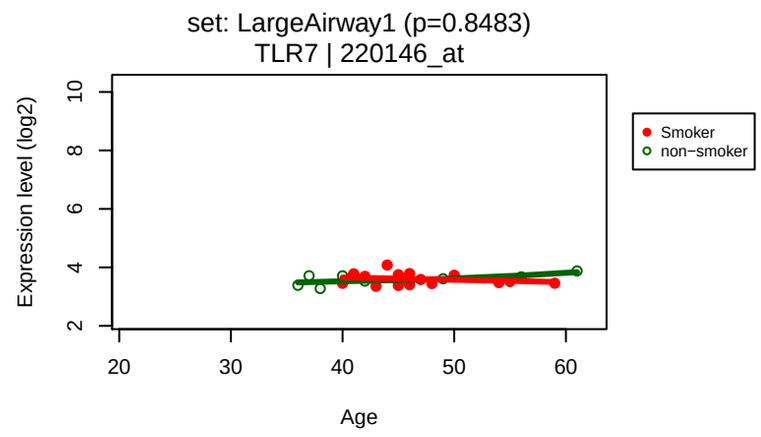

Gene expression of smokers and nonsmokers
gene: TLR7 | 220146_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.8142
p=0.6705
p=0.6611
p=0.8483
p=0.5625
p=0.0671
p=0.0189
p=0.6198
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

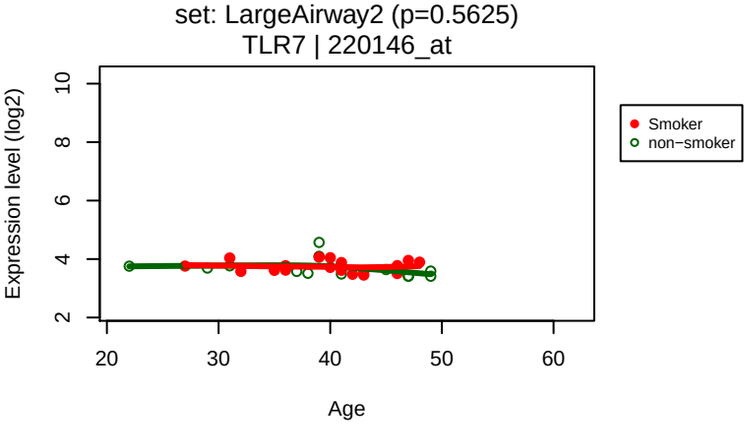

Gene expression ratio (smokers/nonsmokers)
gene: TLR7 | 220146_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8142
p=0.6705
p=0.6611
p=0.8483
p=0.5625
p=0.0671
p=0.0189
p=0.6198
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

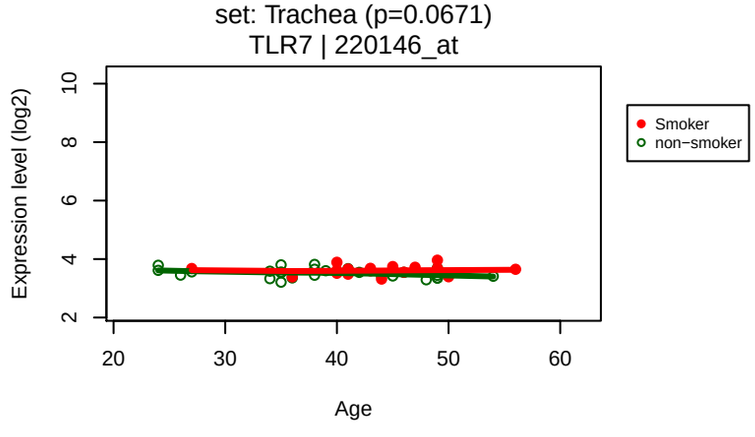

Gene expression ratio in Males
gene: TLR7 | 220146_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8529
p=0.1684
p=0.2755
p=0.5418
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: TLR7 | 220146_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1114
p=0.6038
p=0.2461
p=0.9644
SmallAirway2
LargeAirway2
Trachea
Macrophage2

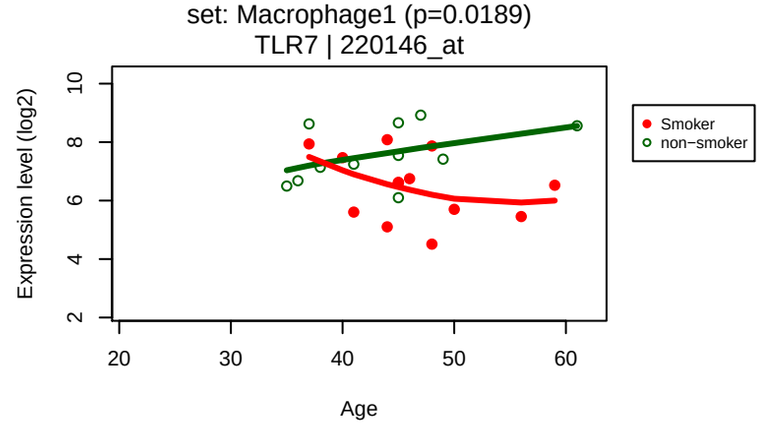

set: SmallAirway1 (p=0.8142)
TLR7 | 220146_at
Expression level (log2)
Age
Smoker
non−smoker

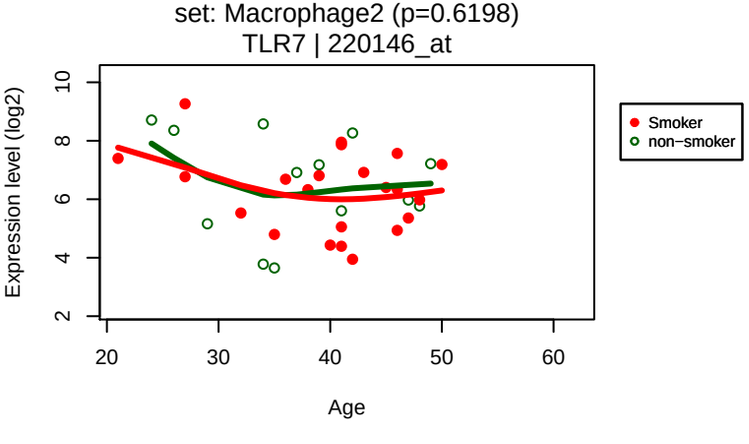

set: SmallAirway2 (p=0.6705)
TLR7 | 220146_at
Expression level (log2)
Age
Smoker
non−smoker

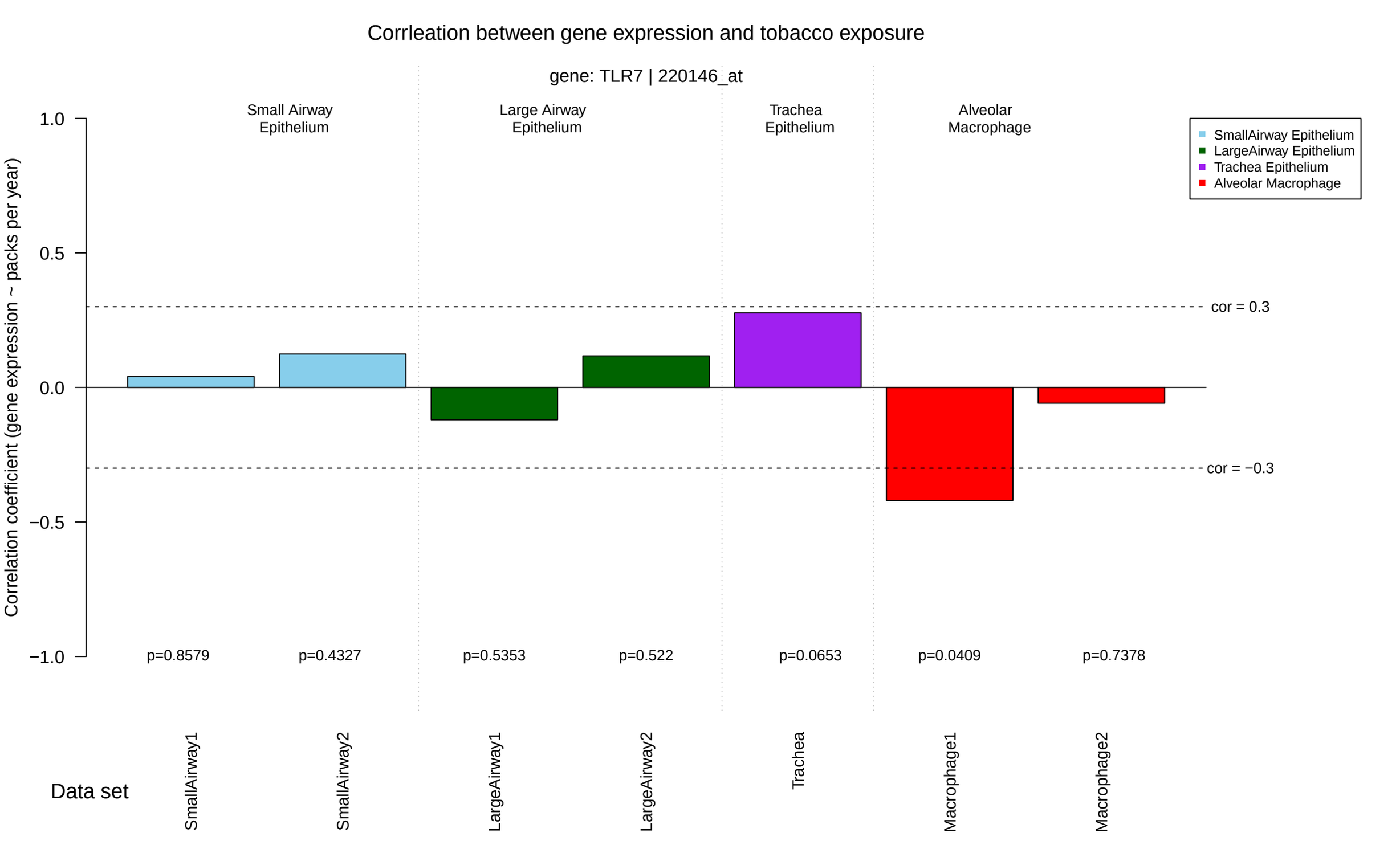

set: LargeAirway1 (p=0.8483)
TLR7 | 220146_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.5625)
TLR7 | 220146_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0671)
TLR7 | 220146_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0189)
TLR7 | 220146_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.6198)
TLR7 | 220146_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: TLR7 | 220146_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.8579
p=0.4327
p=0.5353
p=0.522
p=0.0653
p=0.0409
p=0.7378
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2